



Vivekanand Education Society's Institute of Technology

An Autonomous Institute Affiliated to University of Mumbai

End Semester Examination Winter 2024

Max marks: 60

Duration: 2 hours

Branch: Artificial Intelligence & Data Science (AI&DS)

Semester: III

Name of the Course: Data Structures

QP Code: R24- ADC502_022024-25

Course code: ADC502

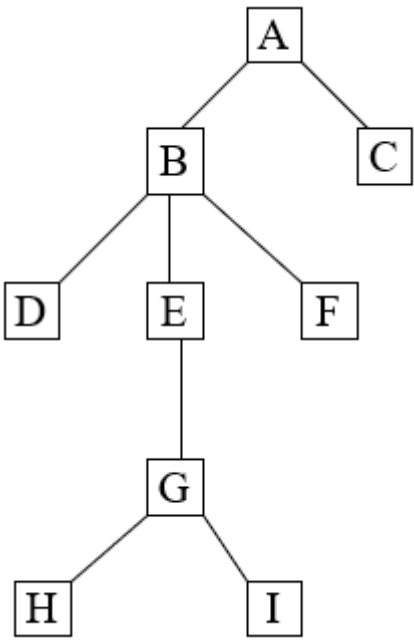
- N.B.** (1) Attempt any three out of the five questions.
(2) Figures to the right indicate full marks.
(3) Assume suitable data if necessary

			Marks	CO	BL
Q.1	(a)	Write a C program to implement a circular queue using arrays.	10	1, 2, 6	L3: Applying
	(b)	Discuss the properties of AVL trees, including their height balance condition. Explain how rotations (single and double) are used to maintain balance during insertions and deletions.	10	3, 4, 6	L2: Understanding
Q.2	(a)	Write functions in C programming language to implement following operations on a singly linked list : deletion (at the beginning, end, and a node having specific data value), and traversal (displaying all nodes).	10	1, 2, 6	L3: Applying
	(b)	Define a B-Tree and explain its structure, including the concepts of order and node capacity. Construct a B-tree of order 5, assuming keys arrive in the following order: 1 12 8 2 25 5 14 28 17 7 52 16 48 68 3 26 29 53 55 45	10	3, 4, 6	L4: Analyzing
Q.3	(a)	Discuss the differences between linked lists and arrays in terms of memory allocation, size flexibility, access time, and ease of insertion/deletion operations. Provide examples to illustrate each point.	10	1, 2, 6	L2: Understanding
	(b)	Write a C program to convert an infix expression to a postfix expression using a stack.	10	1, 2, 6	L3: Applying
Q.4	(a)	Write functions in C programming language to implement Merge Sort. Provide a step-by-step example of Merge Sort on following array. 38, 27, 43, 3, 9, 82, 10, 65	10	5, 6	L3: Applying



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Q.5	(b)	Describe what an Abstract Data Type (ADT) is and explain its importance in data structure design. Provide an example to illustrate your explanation.	5	1, 2, 6	L2: Understanding
	(c)	How singly linked lists can be used in polynomial representation and addition?	5	1, 2, 6	L4: Analyzing
	(a)	With the help of the given diagram explain following terms w.r.to a tree: Root, Siblings, Internal node, External node, Ancestors of a node, Descendant of a node, Depth of a node, Height, Degree of a node, Degree of a tree  <pre> graph TD A[A] --- B[B] A --- C[C] B --- D[D] B --- E[E] B --- F[F] E --- G[G] G --- H[H] G --- I[I] </pre>	10	3, 4, 6	L2: Understanding
	(b)	Build Huffman tree for the given string : hello huffman	5	3, 4, 5, 6	L3: Applying
	(c)	What are Tries, and how are they used in searching for strings?	5	2, 5, 6	L2: Understanding

CO - COURSE OUTCOMES
BL - BLOOM'S TAXONOMY

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